

Distance on the Coordinate Plane

Lesson 9-6

Name: _____

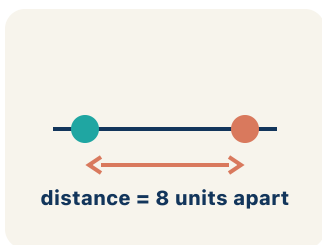
Date: _____

Class: _____

Key Vocabulary

Level 1 support

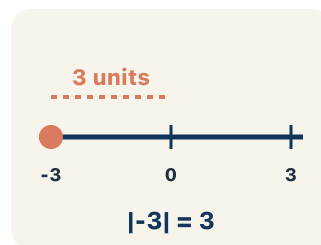
Picture first, then the word, then a plain-language meaning. Say each word out loud.



The distance from -3 to 4 on a number line is $|-3| + |4| = 3 + 4 = 7$ units

Distance

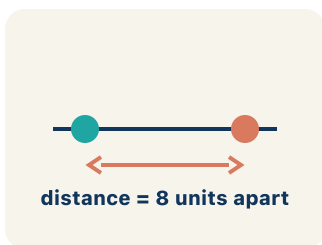
How far apart two points are. It is never negative.



$|-3| = 3$ and $|4| = 4$; to find distance: $|4 - (-3)| = 7$

Absolute value

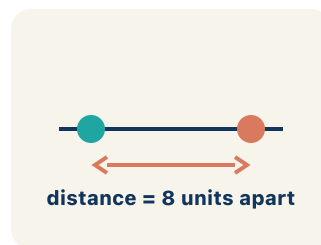
How far a number is from zero.



From $(-2, 3)$ to $(5, 3)$: count from -2 to 5 = 7 units across

Horizontal distance

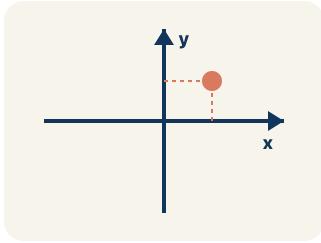
How far apart two points are going left or right.



From $(4, -1)$ to $(4, 6)$: count from -1 to 6 = 7 units up

Vertical distance

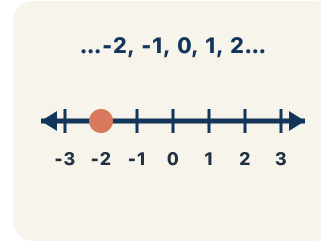
How far apart two points are going up or down.



Two number lines crossing at $(0, 0)$, creating four quadrants

Coordinate plane

A grid with a line going across and a line going up that cross.



Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the distance between two points on the coordinate plane using absolute value.

1 Notice

Captain Vega has two treasure locations on her map: a buried chest at $(-3, 2)$ and a hidden cave at $(4, 2)$.

2 Key idea

I can find the distance between two points on the coordinate plane using absolute value.

3 Words I'll use

Distance

Absolute value

Horizontal distance

Vertical distance

Coordinate plane

Integer



Think About It

- What do the two points $(-3, 2)$ and $(4, 2)$ have in common?
- How can you count the distance between points in the same row?
- How does absolute value help when one point has a negative coordinate?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: What is the distance between $(-4, 3)$ and $(2, 3)$?

- A. 2 units
- B. 4 units
- C. 6 units
- D. 8 units

Step 1 Both points have $y = 3$, so this is a horizontal distance.

Step 2 $|2 - (-4)| = |2 + 4| = 6$ units.

Answer: C. 6 units

 We do — together

Solve this one with your class using the same steps.

Problem: What is the distance between $(5, -2)$ and $(5, 4)$?

- A. 2 units
- B. 4 units
- C. 6 units
- D. 10 units

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A rectangle has vertices at $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$. What is its perimeter?

- A. 14 units
- B. 12 units
- C. 20 units
- D. 8 units

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Two points sit on the same horizontal line: (2, 3) and (7, 3). How can you find the distance between them without counting every square?

Level 1 support

Start here: I can find the distance by looking at the coordinates that ____.

 **Need a hint?**

Sentence starters (optional)

The y-coordinates are the same, so I only compare the ____.

Las coordenadas y son iguales, así que solo comparo las ____.

The distance is ____ because $7 - 2 =$ ____.

La distancia es ____ porque $7 - 2 =$ ____.

Word bank: distance coordinate plane absolute value ordered pair x-axis

Listen for: Student finds the distance by subtracting the x-coordinates ($7 - 2 = 5$) since the points share a horizontal line.

Level 2

Push students to explain why we can use absolute value to find distance even if the points were at negative coordinates.

- Absolute value helps find distance because ____.
- Distance is always positive because ____.

EXPLORE

One point is at $(-3, 1)$ and another is at $(4, 1)$. How does absolute value help you find the distance across zero?

Level 1 support

Start here: Absolute value helps me find the distance across zero because ____.

 **Need a hint?**

Sentence starters (optional)

The distance is $|4| + |-3|$, which equals ____.

La distancia es $|4| + |-3|$, que es ____.

I add the absolute values because the points are on ____ sides of zero.

Sumo los valores absolutos porque los puntos estan en ____ lados del cero.

Word bank: distance absolute value coordinate plane ordered pair negative

Listen for: Student adds the absolute values ($3 + 4 = 7$) because the points are on opposite sides of zero.

Level 2

Ask students to compare finding distance when both points are on the same side of zero versus opposite sides.

- When both points are on the same side of zero, I ____ instead of add.
- When they are on opposite sides, I ____ because ____.

CONNECT

Why does distance on the coordinate plane always come out positive, even when the coordinates are negative?

Level 1 support

Start here: Distance is always positive because ____.

 **Need a hint?**

Sentence starters (optional)

Distance measures how ____ apart two points are.

La distancia mide que tan ____ estan dos puntos.

It can never be negative because ____.

Nunca puede ser negativa porque ____.

Word bank: distance absolute value coordinate plane negative ordered pair

Listen for: Student explains distance is a length (how far apart) and absolute value guarantees a non-negative result.

Level 2

Push students to generalize a rule for finding the distance between any two points on the same horizontal or vertical line.

- A rule for same-line distance is ____.
- I know whether to add or subtract by checking ____.

REFLECT

A classmate found the distance from $(-2, 5)$ to $(6, 5)$ by computing $6 - 2 = 4$. What did they do wrong?

Level 1 support

Start here: The mistake was ____, and the correct distance is ____.

 Need a hint?

Sentence starters (optional)

They forgot that -2 is ____, not ____.

Olvidaron que -2 es ____, no ____.

The correct distance is ____ because ____.

La distancia correcta es ____ porque ____.

Word bank: distance absolute value coordinate plane negative ordered pair

Listen for: Student catches that the classmate ignored the negative sign; correct distance is $|6| + |-2| = 8$.

Level 2

Ask students to critique why ignoring a negative coordinate is a common distance error and how to avoid it.

- This error happens because ____.
- A way to avoid it is to always ____.

Write About the Math

The Writing Revolution

I can explain my work using the words distance, absolute value, horizontal distance, and vertical distance.

1. Kernel Sentence subject + verb

Model: Distance is how far apart two points are. It is never negative.

Distancia es qué tan separados están dos puntos. Nunca es negativo.

Write a kernel sentence about distance. Use a subject and a verb.

Escribe una oración base sobre distancia. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Distance matters in math

Distancia importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Distance matters in math because ____.

Distancia importa en matemáticas porque ____.

but
pero

Distance matters in math, but ____.

Distancia importa en matemáticas, pero ____.

so
entonces

Distance matters in math, so ____.

Distancia importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about distance.
Di un hecho verdadero sobre distance.

Distance ____.

Question *Pregunta*

Ask a question about distance.
Haz una pregunta sobre distance.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about distance.
Muestra entusiasmo sobre distance.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with distance.
Dile a un compañero qué hacer con distance.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The y-coordinates are the same, so I only compare the ____.

Las coordenadas y son iguales, así que solo comparo las ____.

The distance is ____ **because** $7 - 2 =$ ____.

La distancia es ____ *porque* $7 - 2 =$ ____.

The distance is $|4| + |-3|$, **which equals** ____.

La distancia es $|4| + |-3|$, *que es* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A point is at $(-3, 2)$ and another at $(-3, -5)$. How far apart are they?

- A. 7
- B. 3
- C. 2
- D. 8

Show your work:

2. Why can you find the distance between $(1, 4)$ and $(6, 4)$ by subtracting x-values?

- A. They have the same y-coordinate, so the segment is horizontal
- B. All points are the same
- C. Distance is always the larger number
- D. You must always add the coordinates

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A rectangle has vertices at $(-4, -2)$, $(3, -2)$, $(3, 5)$, and $(-4, 5)$. Find the length, width, perimeter, and area of the rectangle. Show your work using absolute value.

Sentence starter: The width is $|___ - (___)| = ___ \text{ units}$. The height is $|___ - (___)| = ___ \text{ units}$. The perimeter is $2(___ + ___) = ___ \text{ units}$. The area is $___ \times ___ = ___ \text{ square units}$.

Show your work:

Reflect — Exit Ticket

What is the distance between the points $(3, -2)$ and $(3, 5)$?

- A. 3 units
- B. 5 units
- C. 7 units
- D. 9 units

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** C. 6 units — *Both points have $x = 5$, so this is a vertical distance. $|4 - (-2)| = |4 + 2| = 6$ units.*
2. **You Try (notes):** A. 14 units — *Width = $5 - 1 = 4$. Height = $4 - 1 = 3$. Perimeter = $2(4 + 3) = 14$.*
3. **Try It 1:** A. 7 — *Same x ; distance = $|2 - (-5)| = 7$ units.*
4. **Try It 2:** A. They have the same y -coordinate, so the segment is horizontal — *Equal y -values mean a horizontal segment, so distance = $|6 - 1| = 5$.*
5. **Exit Ticket:** C. 7 units — *Both points share $x = 3$, so this is a vertical distance. $|5 - (-2)| = |5 + 2| = 7$ units.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Distance is how far apart two points are. It is never negative.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).